

HAT-P-36b

R-1246

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_220528 · created 2026-07-31T11:27:09+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459727.741 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.134 +/- 0.013
Transit depth (Rp/Rs)^2	1.79 +/- 0.34 [%]
Orbital Inclination (inc)	84.06 +/- 2.19
Airmass coefficient 1 (a1)	1.044 +/- 0.016
Airmass coefficient 2 (a2)	-0.034 +/- 0.0095
Scatter in the residuals of the lightcurve fit is	1.51 %
Best Comparison Star	#3 - [30, 97]
Optimal Method	PSF photometry
Transit Duration (day)	0.0889 +/- 0.0075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.134 ± 0.013 vs 0.126 (deviation 0.53σ)
Duration fit vs published	0.0889 d vs 0.0925 d (deviation 0.41σ)
Mid-time O–C	-0.8 min
Depth SNR	8.9
Residual scatter	1.51 %

Light curve

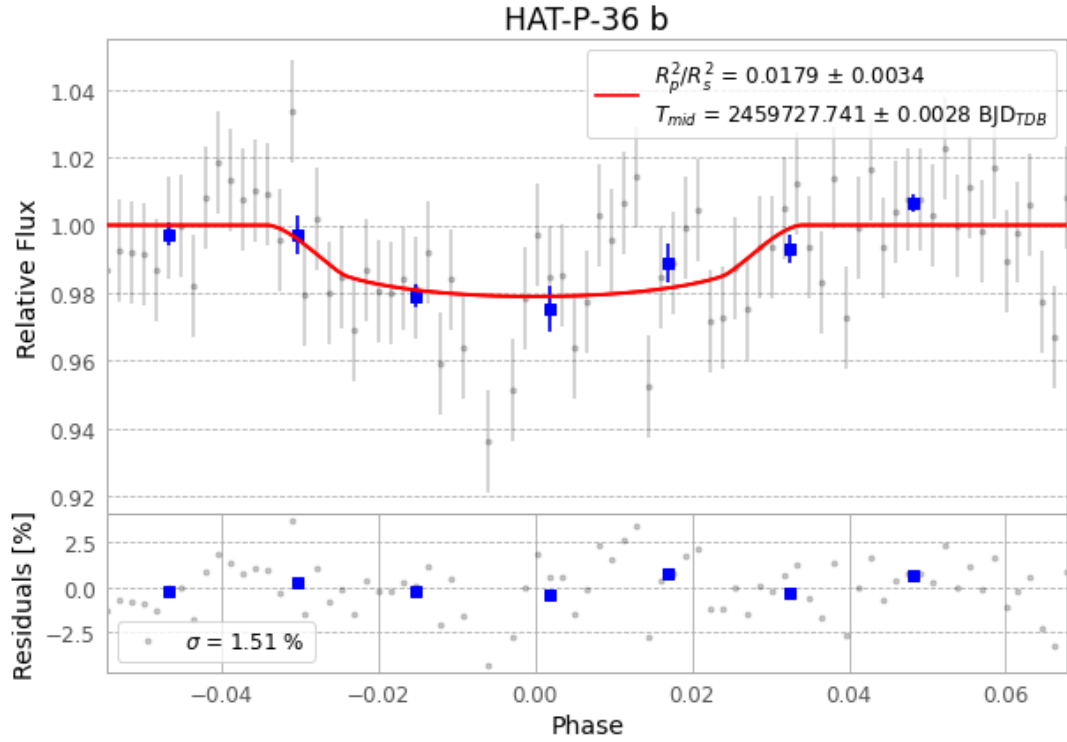


Figure 1 — FinalLightCurve_HAT-P-36 b_2022-05-27.png (SHA-256 464fdecd2290af93...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [300, 164], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 ca43a877c6ae3509...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

79 FITS frames (52 MB), HATP-36220528040013.FITS → HATP-36220528075412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-220528.log (b4a10c7ac94b...), FinalLightCurve_HAT-P-36 b_2022-05-27.png (464fdecd2290...), FinalLightCurve_HAT-P-36 b_2022-05-27.csv (6aed3609c3a7...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 11:27Z.